

Testing Hypotheses Regarding Covariance and Correlation Matrices with the R Package `CovCorTest`

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Abstract In addition to the commonly analyzed measures of location, measures of dispersion and dependence, such as variance and correlation, provide valuable information. Consequently, they play a crucial role in multivariate statistics, which leads to tests regarding covariance and correlation matrices. Furthermore, the structure of these matrices also leads to important hypotheses of interest, as it contains substantial information about the underlying model. In fact, assumptions regarding the structure of covariance and correlation matrices are often fundamental in statistical modeling and testing. In this context, semi-parametric settings with minimal distributional assumptions and very general hypotheses are essential for enabling manifold usage. The freely available package `CovCorTest` provides suitable tests that address all issues mentioned above, using bootstrap and related techniques to achieve good performance, particularly in small samples. Additionally, the package offers flexible specification options for the hypotheses under investigation in two central tests, accommodating users with varying levels of expertise. This paper also illustrates the application of `CovCorTest` for various issues using multiple real-world examples.

1 Introduction

Tests for measures of location are widely used in the analysis of multivariate data, such as those arising from repeated-measures designs or multiple measurements on the same subject. While expectation vectors are a classical choice for such analyses (Konietschke et al., 2021), alternative parameters such as quantiles (Baumeister et al., 2024), relative effects (Rubarth et al., 2022; Dobler et al., 2020) and others have been proposed. In addition to location parameters, measures of dispersion contain substantial information and are therefore of central importance in multivariate settings. In particular, covariance and correlation matrices play a key role as excellent parameters for analyses for several reasons:

- They are comparatively easy to define and understand.
- They quantify the relationships between the components, which most other parameters do not take into account.
- In statistical modeling, covariance and correlation matrices and their structures (e.g. autoregressive or compound symmetry) are essential for understanding and describing the underlying processes.
- Conditions on the covariance or correlation matrix, such as homogeneity or a special structure/pattern, are among the most frequent assumptions of statistical methods.
- Finite second moments and, therefore, the existence of variances are usually presumed and sufficient to ensure consistency of estimators for the unknown covariance or correlation matrices.

The R package `CovCorTest` provides a unified and flexible framework for testing hypotheses about covariance and correlation matrices in multivariate data. In contrast to many existing implementations, it supports general linear hypotheses, requires only minimal distributional assumptions and offers a consistent user interface across a wide range of testing scenarios. The package is designed for applied statisticians and data analysts working with multivariate data, as well as for methodological researchers who require flexible hypothesis specification beyond standard settings.

Due to their importance, a variety of statistical procedures have been developed to test hypotheses concerning covariance matrices. Unfortunately, many existing methods rely on restrictive distributional assumptions, such as normality in Box (1953) and Zhong et al. (2017), or specific conditions on the characteristic function in Gupta and Xu (2006), which limit their applicability. Furthermore, several approaches are limited to specific settings, for example, two-group settings as in Sakaori (2002) or Omelka and Pauly (2012). Moreover, they typically allow the investigation of only one or two concrete hypotheses, such as the equality of covariance matrices (e.g., Yang and DeGruttola, 2012; Zhu et al., 2002) or equality of two correlation matrices (e.g., Jennrich, 1970).

Recent methodological developments address several of these limitations by formulating covariance and correlation hypotheses within a common semi-parametric framework. In particular, the approaches introduced in Sattler et al. (2022) and Sattler and Pauly (2024) allow for a broad class of

hypotheses while requiring only weak distributional assumptions and maintaining good small-sample performance through resampling-based inference procedures. These developments also include combined tests for the simultaneous assessment of covariance and correlation matrices.

Structural hypotheses of covariance and correlation matrices are also of practical relevance, as they provide meaningful insights into the underlying model and frequently serve as model assumptions. Examples include compound symmetry, Toeplitz, and autoregressive structures. However, existing tests for such structures are often available only for selected cases, such as compound symmetry (Winer et al., 1991; Votaw, 1948) or first-order autoregressive structures (McKeown and Johnson, 1996), or focus exclusively on either covariance or correlation matrices (Steiger, 1980). Building on the approaches proposed in Sattler et al. (2022) and Sattler and Pauly (2024), these methods were extended in Sattler and Dobler (2026) to cover a wider range of structural hypotheses on covariance and correlation matrices.

The methodological limitations of earlier approaches are naturally reflected in existing software implementations. For instance, R packages such as *mvhtests*, *biotools*, and *heplots* provide classical procedures for testing the equality of two or more covariance matrices, including Box's *M*-test and likelihood-based approaches, typically relying on multivariate normality. Other packages provide tools for more specific covariance testing problems. For example, *covTestR* (Barnard and Young, 2018) implements tests for high-dimensional covariance-matrix problems, including tests for the homogeneity of two or more covariance matrices and one-sample tests of whether a covariance matrix equals a specified target matrix, such as the identity matrix. Further related functionality is available in *psych* (Revelle, 2019), which provides selected tests and diagnostics for correlation matrices, such as tests of whether a correlation matrix is an identity matrix and tests for the equality of two correlation matrices. Thus, existing packages provide useful procedures for specific settings and selected hypotheses, but their scope is typically limited with respect to the range of hypotheses, the considered data settings, or the underlying assumptions.

The R package *CovCorTest* (Sattler and Jedhoeff, 2026) implements the above recent methodological developments and makes them available within a single and consistent software framework. It includes the hypotheses mentioned above, such as the equality of covariance or correlation matrices and identity-type correlation hypotheses, but also goes substantially beyond them. The package supports general hypothesis formulations for covariance and correlation matrices, structural hypotheses with unknown structure parameters, combined covariance-correlation tests, and manually specified hypotheses through suitable hypothesis matrices. Hence, *CovCorTest* provides a coherent workflow for procedures that are otherwise either unavailable in R or available only as isolated special cases.

The package, which is freely available from the Comprehensive R Archive Network (CRAN), provides five main functions—`test_covariance`, `test_correlation`, `test_covariance_structure`, `test_correlation_structure`, and `test_combined`. These functions are designed to follow a uniform interface, facilitating a consistent and user-friendly workflow across different testing scenarios. This uniform design allows users to apply analogous procedures to hypotheses targeting different aspects of the data and different data structures. Together with the coherent methodological framework, this design enhances usability and improves the comparability of results. Furthermore, to support users with varying levels of experience, the covariance and correlation tests provide predefined hypotheses for ease of use, as well as fully customizable hypotheses—specified via a hypothesis matrix and corresponding vector—for advanced applications.

Because all underlying approaches and tests are based on empirical covariance and correlation matrices, our functions should only be used for metric data, although further assumptions, such as a repeated-measures design, are not required.

In the subsequent section, we will introduce the statistical multivariate model, the investigated null hypotheses, and the underlying assumptions. Afterwards, we present the implemented inference procedures, followed by the applications from the R package *CovCorTest* illustrated by some examples using a real-world dataset. Finally, a conclusion and outlook complete the paper.

2 Statistical model and hypotheses

2.1 Data structure and model assumptions

CovCorTest is designed for multivariate data observed in one or more independent groups. Therefore, we consider a general multigroup setting with $a \in \mathbb{N}$ groups, where each group $i = 1, \dots, a$ contains $n_i \in \mathbb{N}$ observations. The groups may be unbalanced and therefore have different sample sizes, while the total number of observations is denoted by $N = \sum_{i=1}^a n_i$.

Then, a general semi-parametric model is used, defined by independent d -dimensional random

vectors

$$\mathbf{X}_{ik} = \boldsymbol{\mu}_i + \boldsymbol{\epsilon}_{ik}, \quad (1)$$

where $\boldsymbol{\mu}_i \in \mathbb{R}^d$ represents the group-specific mean vector, and $\boldsymbol{\epsilon}_{ik} \in \mathbb{R}^d$ denotes the random deviation for subject $k = 1, \dots, n_i$ from the i -th group. Here, splitting the indices allows considering factorial designs and longitudinal data, as demonstrated, e.g., in [Pauly et al. \(2015\)](#). We assume the residuals $\boldsymbol{\epsilon}_{i1}, \dots, \boldsymbol{\epsilon}_{in_i}$ to be centered, $\mathbb{E}(\boldsymbol{\epsilon}_{i1}) = \mathbf{0}_d$, and i.i.d. within each group, while across different groups, only independence is needed, but differing distributions are possible. Further assumptions are finite fourth moments ($\mathbb{E}(\|\boldsymbol{\epsilon}_{i1}\|^4) < \infty$), existence of group-specific covariance $\mathbf{V}_i$ and correlation matrices $\mathbf{R}_i$ and that asymptotically no group vanishes ($\frac{n_i}{N} \rightarrow \kappa_i \in (0, 1]$ for $i = 1, \dots, a$). No parametric distributional assumptions, such as multivariate normality, are imposed. These weak assumptions ensure the broad applicability of the implemented procedures and are sufficient for the consistent estimation of covariance and correlation matrices. For correlation-based tests, the existence of strictly positive variances is additionally required, as well as $d > 1$.

This framework allows CovCorTest to be applied to a wide range of practical settings while maintaining robustness in small and moderate sample sizes.

2.2 Hypotheses on covariance and correlation matrices

For covariance matrices, the upper triangular part including the diagonal is considered (denoted as vech and resulting in a p -dimensional vector), while for correlation matrices only the off-diagonal elements are used (denoted as vech^- and resulting in a p_- -dimensional vector).

For illustration, consider the case $d = 3$. For a covariance matrix and its corresponding correlation matrix

$$\mathbf{V}_1 = \begin{pmatrix} v_{11} & v_{12} & v_{13} \\ v_{12} & v_{22} & v_{23} \\ v_{13} & v_{23} & v_{33} \end{pmatrix}, \quad \mathbf{R}_1 = \begin{pmatrix} 1 & r_{12} & r_{13} \\ r_{12} & 1 & r_{23} \\ r_{13} & r_{23} & 1 \end{pmatrix}, \quad (2)$$

the corresponding vectors are given by

$$\text{vech}(\mathbf{V}_1) = (v_{11}, v_{12}, v_{13}, v_{22}, v_{23}, v_{33})^\top, \quad \text{vech}^-(\mathbf{R}_1) = (r_{12}, r_{13}, r_{23})^\top. \quad (3)$$

Thus, $\text{vech}(\mathbf{V}_1)$ contains all unique entries of the covariance matrix, including the variances on the diagonal, whereas $\text{vech}^-(\mathbf{R}_1)$ contains only the off-diagonal correlations, since the diagonal entries of a correlation matrix are always equal to one and therefore contain no information for the hypothesis tests.

Now, based on the pooled covariance vector $\mathbf{v} = (\text{vech}(\mathbf{V}_1)^\top, \dots, \text{vech}(\mathbf{V}_a)^\top)^\top$, respectively, the pooled correlation vector $\mathbf{r} = (\text{vech}^-(\mathbf{R}_1)^\top, \dots, \text{vech}^-(\mathbf{R}_a)^\top)^\top$, general linear hypotheses can be formulated.

Using adequate hypothesis matrices $\mathbf{C}_v$ and $\mathbf{C}_r$, respectively, and corresponding vectors $\boldsymbol{\zeta}_v$ and $\boldsymbol{\zeta}_r$, respectively, they are given through

$$\mathcal{H}_0^v : \mathbf{C}_v \mathbf{v} = \boldsymbol{\zeta}_v \quad \text{and} \quad \mathcal{H}_0^r : \mathbf{C}_r \mathbf{r} = \boldsymbol{\zeta}_r. \quad (4)$$

Popular hypotheses included in this model, among many others, are

- Equal covariance matrices: $\mathbf{V}_1 = \dots = \mathbf{V}_a$
- Equal traces of covariance matrices: $\text{tr}(\mathbf{V}_1) = \dots = \text{tr}(\mathbf{V}_a)$
- Equal variances: $(\mathbf{V}_1)_{11} = \dots = (\mathbf{V}_1)_{dd}$
- Equal correlation matrices: $\mathbf{R}_1 = \dots = \mathbf{R}_a$
- Equal correlated components: $\text{vech}^-(\mathbf{R}_1)_1 = \dots = \text{vech}^-(\mathbf{R}_1)_{p_-}$
- Uncorrelated components: $\text{vech}^-(\mathbf{R}_1) = \mathbf{0}_{p_-}$
- Trace of the covariance matrix has a special value $\gamma \in \mathbb{R}_+$: $\text{tr}(\mathbf{V}_1) = \gamma$
- Specific covariance matrix $\mathbf{V} \in \mathbb{R}^{d \times d}$: $\mathbf{V}_1 = \mathbf{V}$

These kinds of hypotheses arise both as research questions and as modeling assumptions in applied multivariate analyses. For instance, homogeneity of covariance matrices is a classical assumption in MANOVA-type analyses and discriminant analysis; see, for example, [Friendly and Sigal \(2020\)](#). Hypotheses on correlation matrices are relevant when comparing dependence structures between populations or experimental conditions, for example, in gene expression studies comparing healthy and tumor tissue; see [Caballe et al. \(2018\)](#). Moreover, hypotheses on equal variances or equal traces can be used to compare marginal or total multivariate variability across groups or populations; see

Le Maître and Mitteroecker (2019). Finally, hypotheses of uncorrelated components are closely related to assessing whether the correlation matrix differs from the identity matrix, a common diagnostic step before factor analysis; see Bartlett (1951) and Revelle (2019).

For ease of use, CovCorTest provides a set of predefined hypotheses that can be selected via simple character arguments. At the same time, advanced users may specify custom hypotheses directly through hypothesis matrices, enabling highly flexible and non-standard testing scenarios. For example, using the centering matrix $P_a = I_a - \mathbf{1}_{a \times a}/a$ and the Kronecker product, hypotheses such as equality of covariance or correlation matrices can be expressed in terms of linear constraints on the vectorized parameters, e.g., $(P_a \otimes I_p)v = \mathbf{0}_{ap}$, and similarly for other hypotheses.

Note that the same hypothesis can often be expressed in multiple ways through different choices of C and ζ (see Sattler and Zimmermann, 2024; Sattler and Rosenbaum, 2025b, for a detailed discussion). In the case of $\zeta = \mathbf{0}$, unique projection matrices exist to check the underlying hypotheses, while in general, the considered hypothesis matrices do not have to be square, idempotent or symmetrical.

2.3 Structural hypotheses

In many applications, assumptions on the *structure* of covariance or correlation matrices are of primary interest. For covariance matrices, common structures include

- Compound symmetry
- Diagonality
- First-order autoregressive (AR(1))
- Sphericity
- Toeplitz structure,

see Kincaid (2005) for more details regarding these structures. For correlation matrices, analogous—but typically fewer—structures exist, which often arise as heterogeneous versions of other structures such as heterogeneous autoregressive, heterogeneous compound symmetry, or heterogeneous Toeplitz structures. CovCorTest supports a broad class of such structural hypotheses by expressing them as linear constraints on vectorized covariance or correlation matrices, including both homogeneous and heterogeneous versions of common structures. Except for autoregressive structures and their variants, for which a transformation is necessary, these covariance and correlation structures can be expressed within the above general hypothesis framework using appropriate choices of C and ζ , as shown in Sattler and Dobler (2026). The package additionally implements banded and banded Toeplitz structures. For banded structures, all entries outside a specified bandwidth q are constrained to be zero, so only the diagonal and the first q off-diagonals may contain non-zero entries. Depending on the selected structure, the entries within this band may either remain unrestricted or be further constrained, for example by a Toeplitz pattern.

All supported structures can be tested using dedicated functions with a unified interface, allowing users to assess structural assumptions in a transparent and reproducible manner. Also, structures from a linear covariance structure model, comparable to Anderson (1973) or Szatrowski (1980), can be expressed through a hypothesis matrix and vector. Although these expressions are not unique, the package includes a function `get_hypothesis`, which, for given vectors specifying the model, returns a possible hypothesis matrix C and corresponding vector ζ . This highlights the versatility and generality of the presented model in also capturing a wide range of relevant covariance and correlation structures.

3 Inference

To evaluate the hypotheses formulated in the previous section, we develop inference procedures based on the asymptotic distributions of vectorized covariance and correlation estimators. For our hypotheses regarding covariance matrices, we consider the vectorized empirical group covariances $\hat{v}_i$ as the counterpart of v_i . Stacking these vectors across all a groups yields the pooled vector $\hat{v} = (\hat{v}_1^\top, \dots, \hat{v}_a^\top)^\top$. Then, for developing a statistical test, we use the fact that under our assumptions of Section 2 and the null hypothesis $\mathcal{H}_0^v$, it holds that

$$\sqrt{N}(C_v \hat{v} - \zeta_v) \xrightarrow{D} \mathcal{N}(\mathbf{0}, C_v \Sigma C_v^\top), \quad (5)$$

with the block-diagonal covariance matrix Σ .

In the same way, $\hat{r}_i$ denotes the empirical counterpart of the upper-half-vectorized correlation matrix of group i and $\hat{r}$ is the corresponding pooled vector. Following the results in Sattler and Pauly

(2024) (where the exact form can also be found), there exists a continuous matrix-valued function $M \in \mathbb{R}^{p \times p}$ such that

$$\sqrt{n_i}(\hat{r}_i - r_i) = M(v_i, r_i) \cdot \sqrt{n_i}(\hat{v}_i - v_i) + \mathcal{O}_p(1). \quad (6)$$

Using this result and the previous asymptotic expansion, we obtain the joint asymptotic distribution under $\mathcal{H}_0'$ as

$$\sqrt{N}(C_r \hat{r} - \zeta_r) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, C_r Y C_r^\top), \quad (7)$$

using a block-diagonal matrix based on the transformation matrix $M(v_i, r_i)$ and Σ .

Based on these limit distributions, to test our multivariate hypotheses, we use the ATS (ANOVA-type statistic), initially developed in Brunner et al. (1997), which in the present setting are given by

$$ATS_v = N[C_v \hat{v} - \zeta_v]^\top [C_v \hat{v} - \zeta_v] / \text{tr}(C_v \hat{\Sigma} C_v^\top), \quad (8)$$

and

$$ATS_r = N[C_r \hat{r} - \zeta_r]^\top [C_r \hat{r} - \zeta_r] / \text{tr}(C_r \hat{Y} C_r^\top). \quad (9)$$

Here, for the unknown covariance, we use the estimators $\hat{\Sigma}$ and $\hat{Y}$, respectively, built from the groupwise estimators and a plug-in version of M (see Sattler et al. (2022) and Sattler and Pauly (2024) for the explicit formulas). This quadratic form is essentially the norm of the asymptotically normally distributed vector. A high value, therefore, leads to rejection, while standardization enhances the statistical stability of the results and ensures invariance under scalar multiplications, such as unit changes.

There also exist alternative versions of the ATS, which, for example, allow the approximation through an F -distribution (see, e.g., Brunner et al. (2019) or Thiel et al. (2026)). However, for the statistics defined in (8) and (9), the limiting null distribution can be characterized as a weighted sum of independent chi-square distributed random variables. Although this distribution is generally not available in closed form, this characterization enables valid inference without stringent distributional assumptions.

For both ATS, the asymptotic distribution under the respective null hypothesis is a weighted χ^2 distribution, i.e.,

$$ATS \xrightarrow{\mathcal{D}} \sum_{\ell=1}^m \lambda_\ell B_\ell, \quad B_\ell \stackrel{i.i.d.}{\sim} \chi_1^2, \quad (10)$$

where m denotes the number of rows of the corresponding hypothesis matrix C . The weights λ_ℓ are given by the eigenvalues of the matrices $C_v \Sigma C_v^\top$ and $C_r Y C_r^\top$, respectively, divided by the trace of the corresponding matrix, and therefore reflect the covariance structure of the underlying estimators.

Since no explicit formula is available for this weighted distribution, several approaches can be used to determine the required critical values. One such approach is a Monte-Carlo procedure, which we describe first. For the covariance and correlation statistics, the consistent estimators $\hat{\Sigma}$ and $\hat{Y}$, respectively, yield estimators $\hat{\lambda}_1, \dots, \hat{\lambda}_m$ for the unknown normalized eigenvalues. With these weights, we generate appropriate realizations $B_1, \dots, B_m \stackrel{i.i.d.}{\sim} \chi_1^2$. These realizations and our estimators allow calculating a weighted sum

$$\sum_{\ell=1}^m \hat{\lambda}_\ell B_\ell, \quad (11)$$

which converges in distribution to the above limit distribution. Repeating this procedure many times allows calculating the required Monte-Carlo quantiles, and with an analogous procedure for the correlation for $\alpha \in (0, 1)$, we get asymptotic level- α tests through

$$\begin{aligned} \varphi_v^{MC} &= \mathbb{1}(ATS_v > q_{v,1-\alpha}^{MC}), \\ \varphi_r^{MC} &= \mathbb{1}(ATS_r > q_{r,1-\alpha}^{MC}). \end{aligned} \quad (12)$$

Since the covariance estimators and the corresponding estimated eigenvalues have to be calculated only once and all involved random variables are univariate, this procedure is computationally efficient even for large numbers of repetitions or large values of m .

A further technique for obtaining quantiles that requires more computational effort but usually yields better small-sample performance is a bootstrap approach. For hypotheses regarding covariance matrices, the parametric bootstrap mimics the asymptotic distribution of the test statistic by generating bootstrap vectors $Z_{i1}^*, \dots, Z_{in_i}^* \stackrel{i.i.d.}{\sim} \mathcal{N}(\mathbf{0}, \hat{\Sigma}_i)$ for each group, using the covari-

ance estimators $\hat{\Sigma}_i$ obtained from the data. For each group, the empirical covariance matrix of the bootstrap sample is calculated and therewith $\hat{\Sigma}^*$. The bootstrap analogue of the ATS is given through $ATS_v^* = N[C_v \bar{Z}^*]^\top [C_v \bar{Z}^*] / \text{tr}(C_v \hat{\Sigma}^* C_v^\top)$, using the bootstrap means $\bar{Z}_i^* = n_i^{-1} \sum_{k=1}^{n_i} Z_{ik}^*$ and $\bar{Z}^* = (\bar{Z}_1^{*\top}, \dots, \bar{Z}_a^{*\top})^\top$. Now, based on a large number of repetitions, the bootstrap quantile $q_{v,1-\alpha}^*$ is determined.

In a similar way, by replacing $\hat{\Sigma}$ and $\hat{\Sigma}_i^*$ with $\hat{Y}$ and $\hat{Y}_i^*$, the bootstrap quantile for the correlation $q_{r,1-\alpha}^*$ can be determined. This allows formulating asymptotic bootstrap tests for the respective hypothesis and $\alpha \in (0, 1)$ by

$$\begin{aligned} \varphi_v^* &= \mathbb{1}(ATS_v > q_{v,1-\alpha}^*), \\ \varphi_r^* &= \mathbb{1}(ATS_r > q_{r,1-\alpha}^*). \end{aligned} \quad (13)$$

Since for each bootstrap run, all observations have to be generated and all empirical covariance matrices have to be calculated, this approach can be time-consuming, especially for higher dimensions. However, it usually leads to improved finite-sample performance, particularly for small sample sizes; see the simulation studies in [Sattler et al. \(2022\)](#) and [Sattler and Pauly \(2024\)](#).

For the investigation of hypotheses regarding correlations, an additional approach exists to obtain the required quantiles. It is called a Taylor-based Monte-Carlo technique and combines elements of Monte-Carlo and bootstrap, using a refined version of the expansion in equation (6) to simulate the asymptotic distribution directly. Further details are provided in [Sattler and Pauly \(2024\)](#). Since for this approach, only a single observation per group is generated, and standardization is based on the trace term in (9), this results in a computationally efficient procedure with good finite-sample performance.

In CovCorTest, a common interface is used, where all three inference strategies—Monte-Carlo, parametric bootstrap, and Taylor-based Monte-Carlo—can be chosen. This design allows users to explicitly select the inference method depending on computational constraints, sample size considerations, or desired accuracy, while keeping the specification of hypotheses unchanged.

Remark. Despite their high flexibility, the linear hypotheses introduced in (4) do not allow the formulation of all relevant structures of covariance and correlation matrices. In particular, the autoregressive structure and all of its versions include some proportionality between the matrix components that cannot be expressed through linear constraints on the vectorized covariance or correlation matrices. To accommodate such nonlinear structures, suitable transformations of the parameter vectors are considered. Specifically, let f denote a differentiable transformation of the vectorized covariance parameters and g the corresponding transformation for correlation parameters. This leads to hypotheses of the form $\mathcal{H}_0^{v,s} : \tilde{C}_v f(v) = \tilde{\zeta}_v$ or $\mathcal{H}_0^{r,s} : \tilde{C}_r g(r) = \tilde{\zeta}_r$. By replacing the original parameter vectors with their transformed empirical counterparts and incorporating the corresponding Jacobian matrix, the above-mentioned tests can be adapted accordingly. A detailed discussion of this approach, along with a useful transformation, referred to as the subdiagonal-mean-ratio, is provided in [Sattler and Dobler \(2026\)](#).

Although these tests are highly versatile due to weak distributional assumptions and a broad class of admissible hypotheses, they do not by themselves provide insight into the specific reasons for rejecting a null hypothesis. In particular, a rejection does not directly reveal whether group differences are driven by changes in marginal variances, in dependence structures, or in both components. Since correlation matrices reflect dependency structures but contain no information about the variances of individual components, whereas covariance matrices include variances but lack a standardized measure of dependence, neither parameter is sufficient on its own.

To address this limitation, [Sattler and Pauly \(2024\)](#) propose a combined test for simultaneously assessing group differences in both covariance and correlation structures. This approach is based on the concept of multiple contrast testing (see, e.g., [Bretz et al., 2001](#)) and is applicable to two-group comparisons ($a = 2$). In addition to testing the null hypothesis of equal covariance matrices, the procedure allows identifying whether detected differences are attributable to marginal variances, to dependence structures, or to both components.

In CovCorTest, this combined testing procedure is implemented in a dedicated function that automates the construction of the test statistic, the simulation-based calibration of critical values, and the decomposition of the global decision into variance- and correlation-specific components. The combined test is based on a vector-valued test statistic

$$T = \begin{pmatrix} T_1 \\ \vdots \\ T_p \end{pmatrix} = \sqrt{N} \left(\begin{pmatrix} \hat{V}_{1,11} \\ \hat{V}_{1,22} \\ \vdots \\ \hat{V}_{1,dd} \\ \hat{\mathbf{r}}_1 \end{pmatrix} - \begin{pmatrix} \hat{V}_{2,11} \\ \hat{V}_{2,22} \\ \vdots \\ \hat{V}_{2,dd} \\ \hat{\mathbf{r}}_2 \end{pmatrix} \right), \quad (14)$$

where the first d components correspond to differences in marginal variances and the remaining components capture differences in correlation structures. Under the null hypothesis $\mathcal{H}_0^v : \mathbf{V}_1 = \mathbf{V}_2$, the statistic T converges in distribution to a multivariate normal distribution. The explicit form of the corresponding asymptotic covariance matrix is provided in [Sattler and Pauly \(2024\)](#).

To obtain critical values, a simulation-based procedure related to the Taylor-based Monte-Carlo approach is employed to approximate the limiting distribution. Specifically, for both groups $i = 1, 2$, independent realizations

$$\Lambda_i \sim \mathcal{N}(\mathbf{0}, \hat{\Sigma}_i), \quad (15)$$

are generated and subsequently transformed into Λ_i^{Tay} . Based on these transformed quantities, a simulated version of the test statistic, denoted by T^{Tay} , is constructed. Repeating this procedure $B \in \mathbb{N}$ times yields empirical component-wise β -quantiles $q_{\ell, \beta}$. The local significance level $\tilde{\beta}$ is calibrated to control the family-wise type-I error rate at level α using the approach of [Munko et al. \(2024\)](#), which assigns equal local error probabilities across all components.

An asymptotic level- α test of the null hypothesis of equal covariance matrices is then defined by

$$\max \left[\max_{\ell=1, \dots, d} \left(1 - \mathbb{1} \left(q_{\ell, \tilde{\beta}/2} \leq T_\ell \leq q_{\ell, 1-\tilde{\beta}/2} \right) \right), \max_{\ell=d+1, \dots, p} \left(1 - \mathbb{1} \left(q_{\ell, \tilde{\beta}/2} \leq T_\ell \leq q_{\ell, 1-\tilde{\beta}/2} \right) \right) \right]. \quad (16)$$

The components responsible for rejection provide additional interpretative information. Rejection in the first group of components indicates differences in marginal variances, whereas rejection in the second group indicates differences in correlation structures. From a software perspective, this decomposition is reflected in the returned output of `CovCorTest`. Specifically, the implementation provides three p -values: a variance-specific p -value, a correlation-specific p -value, and a global p -value defined as the minimum of the two component-wise p -values. This structure allows users to assess both the overall test decision and interpretable component-wise results within a single, coherent workflow.

From a reproducibility standpoint, all inference procedures implemented in `CovCorTest` are fully deterministic conditional on the state of the random number generator. When a random seed is set via `set.seed()`, Monte-Carlo, bootstrap, and Taylor-based Monte-Carlo methods yield reproducible results across repeated runs. The default values for the number of repetitions follow methodological recommendations, while they remain user-accessible.

4 Software and examples

In this section, we analyze datasets under different hypotheses to illustrate the usage of the testing procedures implemented in the `CovCorTest` package. To this end, we first explain the general syntax of the functions, thereby facilitating their application to a real-world dataset. Since most of the implemented tests follow a similar construction and share a common interface, their application is largely consistent across different testing scenarios. We therefore begin with a general overview of the syntax and then highlight the main differences between the functions. In particular, `test_combined` differs from the other functions because it is designed for a more specific hypothesis and setting, resulting in fewer user-specified options.

4.1 Syntax

The implemented test functions follow a common workflow. First, the user provides the data in one of the supported input formats. Second, the hypothesis or structure of interest is specified. Third, the inference method used to approximate the critical value or p -value is chosen. The following schematic call illustrates this workflow for testing equality of covariance matrices across three groups:

```
X <- list(group1, group2, group3)
nv <- vapply(X, nrow, integer(1))
```

```

res <- test_covariance(
  X = X,
  nv = nv,
  hypothesis = "equal",
  method = "MC",
  repetitions = 1000
)

```

Here, the data object `X` contains the group-specific data matrices or data frames, `nv` contains the corresponding group sample sizes, `hypothesis = "equal"` specifies the null hypothesis of equal covariance matrices, `method = "MC"` selects the Monte-Carlo approximation, and `repetitions = 1000` determines the number of Monte-Carlo repetitions. The same general structure is used for the other test functions, with minor differences depending on whether a covariance matrix, a correlation matrix, a structure, or the combined covariance-correlation hypothesis is tested. The most important arguments are described below.

- **X:** The dataset for which the test should be conducted. Two input formats are supported. First, `X` may be a list, where each list element is a numeric matrix or numeric data frame corresponding to one group. In this format, observations are stored in rows and variables in columns. Hence, all list elements must have the same number of columns, while the number of rows may differ between groups. If `nv` is omitted for list input with multiple groups, the group sample sizes are inferred from the numbers of rows of the list elements. Second, `X` may be a single numeric matrix or numeric data frame containing all groups, again with observations in rows and variables in columns. In this case, the observations have to be ordered by group, and the accompanying vector `nv` specifies the group sample sizes. For single-group tests, including covariance and correlation structure tests, `nv` is not required.
- **hypothesis:** For increased user-friendliness, predefined hypotheses can be selected in `test_covariance`, `test_correlation`, `test_covariance_structure` and `test_correlation_structure`. The available options depend on the respective test function:

Covariance matrix: Here, in the case of one group, the options are "equal" (denoting equal variances between the components), "given-trace", "given-matrix" and "uncorrelated". In the case of multiple groups, the predefined hypotheses are "equal" (now denoting equal covariance matrices in the different groups), "equal-trace" and "equal-diagonals".

Correlation matrix: For one group, the hypotheses are "equal-correlated", denoting equal correlation between the components and "uncorrelated", which means that all components have a correlation of zero. In the case of multiple groups, the only hypothesis is "equal-correlated", which here denotes the equality of correlation matrices.

Covariance structure: For covariance structures, the selectable hypotheses are autoregressive ("autoregressive" or "ar"), first-order autoregressive ("fo-autoregressive" or "fo-ar"), diagonal ("diagonal" or "diag"), sphericity ("sphericity" or "spher"), compound symmetry ("compoundsymmetry" or "cs"), Toeplitz ("toeplitz" or "toep"), constant off-diagonal ("constant-offdiagonal" or "const-offdiag"), standard Toeplitz ("standard-toeplitz" or "std-toep"), banded ("banded" or "band"), and banded Toeplitz ("banded-toeplitz" or "band-toep"). For the banded structures, the additional argument `bandwidth` specifies the number q , $1 \leq q \leq d - 2$, of off-diagonals that are allowed to be non-zero.

Correlation structure: The predefined structures of the correlation matrices are heterogeneous autoregressive ("hautoregressive" or "har"), heterogeneous Toeplitz ("htoeplitz" or "htoep"), heterogeneous compound symmetry ("hcompoundsymmetry" or "hcs"), diagonal ("diagonal" or "diag"), banded ("banded" or "band"), and banded Toeplitz ("banded-toeplitz" or "band-toep"). For the latter two, the same bandwidth argument is used.

- For `test_covariance` and `test_correlation`, more experienced users can specify custom hypotheses through a hypothesis matrix together with a corresponding vector, as introduced in [Sattler et al. \(2022\)](#) or [Sattler and Pauly \(2024\)](#), especially for less standard hypotheses, which increases flexibility. Both refer to the corresponding kind of upper triangular vectorization, and unlike in other procedures, the hypothesis matrix here need be neither a projection matrix nor square. This especially allows structures from the linear covariance structure model and the linear correlation structure model by using the function `get_hypothesis`.
- The function `test_combined`, which jointly assesses the equality of covariance and correlation matrices, is restricted to one specific hypothesis. Consequently, this argument is not available in that context.

- *AM*: Specifies whether the internal calculations are performed using an alternative hypothesis matrix. With $AM = 1$, a companion matrix of the original hypothesis matrix is used. This matrix has fewer rows but yields the same value of the ATS and can therefore reduce the computational effort. The companion matrix is constructed using the approach implemented in [Sattler and Rosenbaum \(2025a\)](#).
- *method*: Specifies the method by which the quantile for the test is calculated. With *BT*, the quantiles are determined through a parametric bootstrap and with *MC* through a Monte-Carlo technique, both based on [Sattler et al. \(2022\)](#) and [Sattler and Pauly \(2024\)](#). In the latter, there is even another method, a Taylor-based Monte-Carlo approach, which leads to the option *Tay* for `test_correlation`. Again, the combined test works only for a Taylor-based Monte-Carlo approach, which makes any options here unnecessary.
- *repetitions*: The number of Monte-Carlo or bootstrap repetitions used in the approximation of the quantiles. This number substantially influences both the test's performance and its computation time. Since a small number of repetitions increases the influence of the randomness on the results, a predefined value of 1000 is a good trade-off, and substantially smaller numbers are not recommended.

The conducted test returns an object containing both the numerical test result and additional information on the specified hypothesis, the chosen method, and the data structure. The most relevant output components are described in the following subsection.

4.2 Output

After conducting a test, the result is stored in an object, such as `res` in the schematic example above. For the output, we distinguish between the combined test and all other implemented test functions.

For the latter functions, the returned object consists of several components. Some of them provide descriptive information about the conducted test, such as the investigated null hypothesis, the method used to approximate the critical value or p-value, the number of repetitions, and the vector containing the group sample sizes. If the hypothesis was chosen from the predefined options, it is reported through the corresponding label. In addition, the returned object contains the hypothesis matrix C and vector ζ used in the calculation, as well as the corresponding original quantities C_{original} and ζ_{original} , which contain the hypothesis before the optional companion-matrix transformation.

Moreover, the estimated covariance matrix used for the test statistic, the value of the test statistic, and the resulting p-value are included in the output.

The print method gives a short and compact summary of the conducted test. For instance, in the case of a correlation test, it reports that a correlation test was performed, the number of groups, the tested hypothesis, for example, equality of correlation matrices in a multi-group setting, the value of the test statistic, and the resulting p-value. Thus, the user can directly identify the tested hypothesis and interpret the result based on the reported p-value. The printed output focuses on the quantities usually needed for interpreting the result, while the full returned object contains additional information that can be inspected if required. In the examples below, we therefore display the printed output and interpret the resulting p-values in the context of the corresponding hypotheses.

For the combined test, the hypothesis, the number of groups, and the approximation method are fixed, since this function is designed for one specific two-group hypothesis. Moreover, due to the Taylor-based Monte-Carlo approach used in this test, no covariance estimator of the test statistic is needed and therefore none is returned. In addition, the test statistic is vector-valued and non-standardized, making its components difficult to compare directly. Since conclusions based only on the numerical value of this test statistic are usually not meaningful, these values are not printed in order to avoid misinterpretation.

Instead, the combined test returns and prints three p-values: `pvalue_Variances` for the local hypothesis of equal variances, `pvalue_Correlations` for the local hypothesis of equal correlations, and `pvalue_Total` for the global hypothesis combining the variance and correlation parts. These p-values allow the user to distinguish whether possible differences are mainly related to the marginal variances, to the standardized dependence structure, or to the combined covariance-correlation hypothesis.

4.3 Examples

To demonstrate the usage of the main functions and to illustrate typical outputs, we present several example analyses based on real-world data. For this purpose, we use the dataset EEG, which includes multivariate measurements from a neurological study conducted at the University Clinic of Salzburg, Department of Neurology. This dataset is part of the [MANOVA.RM](#) package ([Friedrich et al., 2019](#))

and contains $N = 160$ patients from both genders, with three different diagnoses of impairments (subjective cognitive complaints (SCC), mild cognitive impairment (MCI), and Alzheimer's disease (AD)). Thereby, two neurological parameters (z-score of the brain rate and Hjorth complexity) were measured at three different locations of the head: temporal, frontal, and central. This design results in a multivariate factorial structure with multiple response variables and multiple independent groups and was further described in [Bathke et al. \(2018\)](#). In the following sections, we apply the core testing procedures from `CovCorTest` to this dataset, showcasing the practical implementation and interpretation of results.

Covariance matrix test

We first consider the equality of covariance matrices across diagnostic groups. Due to the group sample sizes, we restrict this example to female patients with the three diagnoses SCC, MCI, and AD, leading to $a = 3$ unbalanced groups. The six selected variables correspond to the two neurological parameters measured at the three locations. The data are stored in the list `X`, where each list element contains the observations from one diagnosis group, and `nv` contains the corresponding group sample sizes.

The null hypothesis is

$$H_0 : V_{SCC} = V_{MCI} = V_{AD}. \quad (17)$$

We first test this hypothesis using the predefined option `hypothesis = "equal"`. To compare the available approximation methods, the same test is performed once with the Monte-Carlo approach and once with the parametric bootstrap. Afterwards, we specify the same null hypothesis manually through a hypothesis matrix and a corresponding vector.

```
data("EEGwide", package = "MANOVA.RM")
vars <- colnames(EEGwide)[1:6]
SCCFemale <- EEGwide[EEGwide$sex == "W" & EEGwide$diagnosis == "SCC", vars]
MCIFemale <- EEGwide[EEGwide$sex == "W" & EEGwide$diagnosis == "MCI", vars]
ADFemale <- EEGwide[EEGwide$sex == "W" & EEGwide$diagnosis == "AD", vars]

X <- list(SCCFemale, MCIFemale, ADFemale)
nv <- unlist(lapply(X, nrow))

set.seed(123)
test_covariance(X = X, nv = nv, hypothesis = "equal",
               method = "MC", repetitions = 1000)

#>
#>
#>          Covariance Test
#>          3 groups
#>
#> Hypothesis:          equal
#> Teststatistic value:    2.6042
#> p-value:             p = 0.02600
#>
#> p-value computed using Monte-Carlo-technique with B=1000 repetitions

set.seed(123)
test_covariance(X = X, nv = nv, hypothesis = "equal",
               method = "BT", repetitions = 1000)

#>
#>
#>          Covariance Test
#>          3 groups
#>
#> Hypothesis:          equal
#> Teststatistic value:    2.6042
#> p-value:             p = 0.02300
#>
#> p-value computed using Bootstrap with B=1000 repetitions
```

For the manual specification of the hypothesis matrix, P_3 , the three-dimensional centering matrix, is used. This is the standard projection matrix for comparing three groups. Since in this example each group is represented by a 21-dimensional covariance vector, the hypothesis matrix is constructed as the Kronecker product of the centering matrix and the corresponding identity matrix. The corresponding hypothesis vector is the zero vector. For further details on this type of hypothesis formulation, see [Sattler and Pauly \(2018\)](#). This manual formulation is included to illustrate that the same hypothesis can either be selected through a predefined option or specified explicitly by the user.

```
d <- 6
p <- d * (d + 1) / 2
P3 <- diag(3) - matrix(1 / 3, 3, 3)

Hypothesismatrix <- P3 %x% diag(1, p, p)
Hypothesisvector <- rep(0, 3 * p)

set.seed(123)
test_covariance(
  X = X, nv = nv, C = Hypothesismatrix, Xi = Hypothesisvector,
  method = "BT", repetitions = 1000
)

#>
#>
#>          Covariance Test
#>          3 groups
#>
#> Hypothesis:          C * v = Xi
#> Teststatistic value:    2.6042
#> p-value:             p = 0.02300
#>
#> p-value computed using Bootstrap with B=1000 repetitions
```

The two tests based on the predefined hypothesis lead to the same value of the test statistic, since the test statistic itself does not depend on the chosen approximation method. The p-values differ only slightly because the critical values are approximated by different simulation-based procedures. With p-values of 0.026 and 0.023, respectively, the null hypothesis of equal covariance matrices across the three diagnosis groups is rejected at the usual 5% level. The manually specified hypothesis leads to the same test statistic and p-value as the predefined formulation, confirming that both calls test the same null hypothesis.

Correlation matrix test

After comparing the covariance matrices for women with these diagnoses, the correlation matrices are also investigated for equality. In addition to the parametric bootstrap, the test is also conducted with the Taylor-based Monte-Carlo approach, specifically developed for testing vectorized correlation matrices.

```
set.seed(123)
test_correlation(X = X, nv = nv, hypothesis = "equal-correlated",
  method = "BT", repetitions = 1000)

#>
#>
#>          Correlation Test
#>          3 groups
#>
#> Hypothesis:          equal-correlated
#> Teststatistic value:    0.7432
#> p-value:             p = 0.6060
#>
#> p-value computed using Bootstrap with B=1000 repetitions

set.seed(123)
test_correlation(X = X, nv = nv, hypothesis = "equal-correlated",
  method = "Tay", repetitions = 1000)
```

```
#>
#>
#>           Correlation Test
#>           3 groups
#>
#> Hypothesis:           equal-correlated
#> Teststatistic value:   0.7432
#> p-value:              p = 0.6150
#>
#> p-value computed using Taylor-based Monte-Carlo-approach with B=1000 repetitions
```

Here, the p-values are considerably higher than for the comparison of covariance matrices. This is plausible, since equality of covariance matrices implies equality of the corresponding correlation matrices, whereas the converse does not hold in general. Hence, the rejection of equal covariance matrices may be driven by differences in marginal variances rather than by differences in the standardized dependence structure. Therefore, the hypothesis of equal correlation matrices is not rejected at the usual α levels.

Combined test

Instead of testing equality of covariance matrices and equality of correlation matrices for two groups separately, the combined test from [Sattler and Pauly \(2024\)](#) allows both aspects to be investigated within one procedure. This test decomposes the comparison into variance-related and correlation-related components. It can therefore be used to assess whether differences between two covariance matrices are mainly driven by differences in the marginal variances, by differences in the standardized dependence structure, or by both. In this example, we compare male patients with AD and MCI. The six selected variables are the same as in the previous examples and are stored in the list `X`, while `nv` contains the two group sample sizes.

```
data("EEGwide", package = "MANOVA.RM")
vars <- colnames(EEGwide)[1:6]

X <- list(
  EEGwide[EEGwide$sex == "M" & EEGwide$diagnosis == "AD", vars],
  EEGwide[EEGwide$sex == "M" & EEGwide$diagnosis == "MCI", vars]
)
nv <- unlist(lapply(X, nrow))

set.seed(123)
test_combined(X, nv)

#>
#>
#>           Combined Test
#>
#> p-value-Variances:      p = 0.3880
#> p-value-Correlations:   p = 0.02600
#> p-value-Total:         p = 0.02600
#>
#> p-values computed using Taylor-based Monte-Carlo-approach with B=1000 repetitions
```

The output contains three p-values. The p-value for the variance part is 0.388 and therefore gives no evidence against equality of the marginal variances. In contrast, the p-value for the correlation part is 0.026, indicating evidence against equality of the correlation matrices at the usual (5%) level. The global p-value is also 0.026, so the combined null hypothesis is rejected. In this example, the detected difference between the two groups is therefore mainly attributable to the standardized dependence structure rather than to the marginal variances.

Structure tests

In this analysis, we want to investigate whether the position of the measuring points influences the measured values. While [MANOVA.RM](#) could investigate a related question regarding mean effects,

it does not assess the covariance or dependence structure between the components. The present approach therefore addresses a different aspect of the data by testing whether the covariance matrix is compatible with a compound symmetry structure. This structure assumes that all variances are equal and that all covariances between different components are equal, as would be expected if the measurement positions were exchangeable with respect to their variability and covariance pattern. If this hypothesis is rejected, this indicates that the variability and covariance pattern depend on the position of the measuring points. Moreover, to check whether the standardized dependencies between the measuring points are equal, we additionally test for a heterogeneous compound symmetry structure of the correlation matrix.

```
X <- as.matrix(EEGwide[EEGwide$sex == "W" & EEGwide$diagnosis == "AD",
  c("brainrate_temporal", "brainrate_frontal", "brainrate_central",
    "complexity_temporal", "complexity_frontal", "complexity_central")])

set.seed(123)
test_covariance_structure(
  X = X, structure = "compoundsymmetry", method = "MC"
)

#>
#>
#>          Covariance Test
#>          one group
#>
#> Hypothesis:          compoundsymmetry
#> Teststatistic value:    1.8809
#> p-value:              p = 0.08800
#>
#> p-value computed using Monte-Carlo-technique with B=1000 repetitions

set.seed(123)
test_correlation_structure(
  X = X, structure = "hcompoundsymmetry", method = "MC"
)

#>
#>
#>          Correlation Test
#>          one group
#>
#> Hypothesis:          hcompoundsymmetry
#> Teststatistic value:    3.5208
#> p-value:              p = 0.01000
#>
#> p-value computed using Monte-Carlo-technique with B=1000 repetitions
```

The covariance structure hypothesis is not rejected at the usual (5%) significance level, since the corresponding p-value is 0.088. Thus, the test does not provide sufficient evidence against a compound symmetry structure of the covariance matrix in this example. In contrast, the heterogeneous compound symmetry structure of the correlation matrix is rejected, with a p-value of 0.010. This indicates that the standardized dependencies between the components are not all equal. In the present setting, this provides evidence that the correlation structure of the measurements depends on the considered components.

We can also test the compound symmetry structure of the covariance matrix using a hypothesis formulation based on `get_hypothesis` and `test_covariance`. The function `get_hypothesis` constructs a hypothesis matrix and a corresponding vector for linear covariance or correlation structure models based on a vector v_0 and a matrix V . The columns of V are the vectorized basis components $v_1, \dots, v_m$ of the model. Hence, admissible parameter vectors are of the form

$$v_0 + \theta_1 v_1 + \dots + \theta_m v_m, \quad (18)$$

with unknown coefficients $\theta_1, \dots, \theta_m$. For the compound symmetry example, v_0 is zero and the two columns of V encode the diagonal and off-diagonal components. With the resulting hypothesis matrix and vector, the test for this structure can be conducted.

```

X <- as.matrix(EEGwide[EEGwide$sex == "W" & EEGwide$diagnosis == "AD",
  c("brainrate_temporal", "brainrate_frontal", "brainrate_central",
    "complexity_temporal", "complexity_frontal", "complexity_central")])

v0 <- rep(0, 21)
v_auxiliary <- c(1, rep(0, 5), 1, rep(0, 4), 1, rep(0, 3),
  1, rep(0, 2), 1, 0, 1)
V <- cbind(v_auxiliary, 1 - v_auxiliary)
h <- get_hypothesis(v0, V)

set.seed(123)
test_covariance(
  X = X, C = h$hypothesis_matrix, Xi = h$hypothesis_vector,
  method = "MC", repetitions = 1000
)

#>
#>
#>          Covariance Test
#>          one group
#>
#> Hypothesis:          C * v = Xi
#> Teststatistic value:    4.0926
#> p-value:              p = 0.006000
#>
#> p-value computed using Monte-Carlo-technique with B=1000 repetitions

```

Although this is based on the same dataset, structure, and seed, the p-value differs from the predefined structure test above. This illustrates that, for the ATS, the test decision can depend on the representation of the null hypothesis, as also discussed in [Sattler and Zimmermann \(2024\)](#). The manual formulation therefore serves two purposes: it demonstrates how users can specify structures through `get_hypothesis`, and it shows that different but equivalent formulations of a hypothesis may lead to different finite-sample behavior for this type of test statistic.

5 Conclusion and outlook

In the presented paper, we illustrate the functionality and scope of the R package `CovCorTest`, which provides a comprehensive framework for hypothesis testing concerning covariance and correlation matrices in multivariate data analysis. It contains tests for both parameters and different hypotheses in a uniform way, which excels through its applicability to a large class of possible hypotheses and fewer distributional assumptions. Apart from multi-group settings, single groups are also allowed, as well as factorial designs. Essentially, it contains functions for five different tests: regarding covariance matrices, correlation matrices, a combined test for equality of covariance and correlation matrices, and tests regarding their corresponding structures. To this end, quantiles, based on various techniques including resampling and Monte-Carlo approaches, are used. Although the Monte-Carlo approach leads to the least time-consuming procedure, a parametric bootstrap for covariances and a Taylor-based Monte-Carlo approach for correlations are preferable for smaller sample sizes.

Future developments will focus on methodological extensions that require additional investigation before they can be recommended as routine software features. In particular, multiple contrast tests for general covariance or correlation hypotheses and extensions of the combined covariance-correlation test to more than two groups lead to a rapidly increasing number of local alternatives. This may result in poor finite-sample performance and difficult interpretation of the resulting local decisions. We therefore regard these extensions as topics for further methodological research rather than as stable features for the present software release. Since the approach proposed in [Sattler et al. \(2026\)](#) allows equivalent or redundant hypotheses to be combined into smaller hypothesis systems, it may provide a promising starting point for addressing this issue. Further possible developments concern usability and computational efficiency.

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